

Model 1: Two Coupled Classical Lorentz Oscillators

Analytical benchmark for PlasMol parallel absorption

PlasMol validation notes

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Abstract

We treat the Au nanoparticle LSPR and the Na atomic transition as two driven, damped harmonic oscillators coupled by a near-field constant κ . The linear system is solved in closed form for the frequency-dependent displacements, yielding an effective molecular polarizability $\alpha_m^{\text{eff}}(\omega) = \mu_m(\omega)/E_{\text{inc}}(\omega)$ that is directly comparable (up to peak normalization) to PlasMol's hybrid Fourier spectrum $A(\omega) \propto -\omega \text{Im}[\mu(\omega)/E_{\text{inc}}(\omega)]$.

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1 Motivation and correspondence to PlasMol

PlasMol’s parallel Fourier path for a nanoparticle–molecule system records the molecular induced dipole $\mu(t)$ under a broadband Meep drive and divides by a vacuum incident field E_{inc} sampled at the molecule site (no NP, no molecule). The published spectrum is

$$A(\omega) \propto -\omega \operatorname{Im} \left[\frac{\mu(\omega)}{E_{\text{inc}}(\omega)} \right], \quad (1)$$

peak-normalized over a user energy window.

Equation (1) is *not* the hybrid extinction of the composite. It is the molecular channel relative to free-space incidence, with nanoparticle scattering retained inside μ . Model 1 targets exactly that observable with the simplest two-level (two oscillator) dynamics.

Target system (defaults in `spectrum.model1.py`).

Subsystem	Parameters
Au LSPR	$\omega_p = 2.384 \text{ eV}$, damping γ_p
Na transition	$\omega_m = 2.109 \text{ eV}$, damping γ_m
Detuning	$\delta = \omega_p - \omega_m \approx 0.275 \text{ eV}$
Coupling	κ (units of energy^2), user-tunable

2 Bare Lorentz oscillators

A single classical Lorentz oscillator with resonance ω_j , damping γ_j , and drive strength f_j obeys

$$\ddot{x}_j + \gamma_j \dot{x}_j + \omega_j^2 x_j = f_j E(t), \quad j \in \{p, m\}. \quad (2)$$

In the frequency domain ($e^{-i\omega t}$ convention, or equivalently replacing $\partial_t \rightarrow -i\omega$ in the Fourier domain used here),

$$\Omega_j^2(\omega) x_j(\omega) = f_j E(\omega), \quad \Omega_j^2(\omega) \equiv \omega_j^2 - \omega^2 - i\gamma_j \omega. \quad (3)$$

The bare polarizability (reduced units where $\mu_j \propto x_j$) is

$$\alpha_j^{(0)}(\omega) = \frac{f_j}{\Omega_j^2(\omega)} = \frac{f_j}{\omega_j^2 - \omega^2 - i\gamma_j \omega}. \quad (4)$$

Absorption of an isolated oscillator is proportional to $-\omega \operatorname{Im} \alpha_j^{(0)}(\omega)$, which is a non-negative Lorentzian-like peak of width set by γ_j .

3 Coupled equations of motion

Near-field coupling between the plasmon and molecular oscillators is encoded as a bilinear term with real coupling constant κ (dimensions of $\text{frequency}^2 / \text{energy}^2$ when ω is in eV):

$$\Omega_p^2(\omega) x_p + \kappa x_m = f_p E_{\text{inc}}, \quad (5)$$

$$\kappa x_p + \Omega_m^2(\omega) x_m = f_m E_{\text{inc}}. \quad (6)$$

In matrix form,

$$\begin{pmatrix} \Omega_p^2 & \kappa \\ \kappa & \Omega_m^2 \end{pmatrix} \begin{pmatrix} x_p \\ x_m \end{pmatrix} = \begin{pmatrix} f_p \\ f_m \end{pmatrix} E_{\text{inc}}. \quad (7)$$

The determinant

$$\Delta(\omega) = \Omega_p^2 \Omega_m^2 - \kappa^2 \quad (8)$$

vanishes at the hybrid poles. Cramer's rule gives the closed-form displacements

$$x_p(\omega) = \frac{\Omega_m^2 f_p - \kappa f_m}{\Delta(\omega)} E_{\text{inc}}, \quad (9)$$

$$x_m(\omega) = \frac{\Omega_p^2 f_m - \kappa f_p}{\Delta(\omega)} E_{\text{inc}}. \quad (10)$$

Connection to a coupling energy g . In the literature one often writes a Hamiltonian coupling $\hbar g(a^\dagger \sigma_- + \text{h.c.})$ or a force constant proportional to g . Near resonance, the level repulsion scale is set by an effective g with $g^2 \sim \kappa$ (order of magnitude, depending on oscillator normalizations). For detuning $\delta = \omega_p - \omega_m$, clear mode hybridization (avoided crossing / Rabi-like doublet in the linear spectrum) typically requires $g \gtrsim |\delta|/2$ and g comparable to the mean linewidth. Weak κ yields a single molecular peak that is shifted and reshaped—the usual weak-coupling / Purcell-type regime.

4 Effective molecular polarizability and absorption

Matching PlasMol's definition, the molecular response relative to the incident field is

$$\alpha_m^{\text{eff}}(\omega) \equiv \frac{\mu_m(\omega)}{E_{\text{inc}}(\omega)} \propto \frac{x_m(\omega)}{E_{\text{inc}}} = \frac{\Omega_p^2 f_m - \kappa f_p}{\Omega_p^2 \Omega_m^2 - \kappa^2}. \quad (11)$$

The model spectrum is

$$A(\omega) = -\omega \text{Im}[\alpha_m^{\text{eff}}(\omega)], \quad (12)$$

subsequently peak-normalized,

$$A_{\text{norm}}(\omega) = \frac{A(\omega)}{\max_{\omega} |A(\omega)|}, \quad (13)$$

with a global sign flip if the dominant extremum is negative (same practical convention as PlasMol's `orient_spectrum_sign`).

Limits.

- $\kappa \rightarrow 0$: $\alpha_m^{\text{eff}} \rightarrow f_m/\Omega_m^2 = \alpha_m^{(0)}$ (bare Na).
- $f_m \rightarrow 0$ but $\kappa \neq 0$: the molecule can still be driven indirectly through the plasmon term $-\kappa f_p/\Delta$, i.e. a pure near-field imprint of the LSPR on μ_m .
- Large κ : hybrid poles split; $A(\omega)$ can show two peaks even though only μ_m is plotted.

5 Hybrid mode frequencies

The undriven poles satisfy $\Delta(\omega) = 0$. A useful algebraic estimate replaces Ω_j^2 by complex constants $z_j = \omega_j^2 - i\gamma_j\omega_j$ evaluated at the bare peaks and solves

$$(z_p - \lambda)(z_m - \lambda) - \kappa^2 = 0 \quad \Rightarrow \quad \lambda_{\pm} = \frac{z_p + z_m}{2} \pm \frac{1}{2} \sqrt{(z_p - z_m)^2 + 4\kappa^2}, \quad (14)$$

then $\omega_{\pm} \approx \sqrt{\lambda_{\pm}}$ (principal branch). The script prints $\omega_{\pm}$ as vertical guides. They are indicative, not exact roots of the full transcendental $\Omega_p^2(\omega)\Omega_m^2(\omega) = \kappa^2$.

In the equal-damping, near-resonant textbook limit one recovers the familiar form

$$\omega_{\pm} \approx \frac{\omega_p + \omega_m}{2} - \frac{i}{4}(\gamma_p + \gamma_m) \pm \frac{1}{2}\sqrt{\left(\delta - \frac{i}{2}(\gamma_p - \gamma_m)\right)^2 + 4g^2}, \quad (15)$$

with g the coupling energy related to κ through the chosen oscillator normalization.

6 What to expect for Au (2.384 eV) + Na (2.109 eV)

1. **Weak κ .** Single peak near ω_m , possibly red- or blue-shifted by a fraction of γ_m . A weak shoulder near ω_p may appear if f_p and κ allow plasmon-mediated driving of x_m .
2. **Moderate κ .** Noticeable shift of the molecular maximum; lineshape becomes asymmetric (Fano-like) when the broad plasmon continuum interferes with the narrow molecular resonance.
3. **Strong κ .** Two hybrid peaks; energy splitting grows with $\sqrt{\kappa}$. Peak-normalization still forces $\max |A| = 1$, so compare *positions and relative heights*, not absolute scale.

Your PlasMol `parallel_abs` result—one dominant peak near the Na energy—is the weak-to-moderate coupling pattern. Model 1 validates the *method* when:

- $\kappa = 0$ recovers the bare Na lineshape used as input;
- increasing κ shifts / splits $A(\omega)$ in the direction predicted by $\omega_{\pm}$;
- PlasMol’s hybrid curve lies in the same qualitative regime as a κ estimated from geometry (see Model 2 for a geometry-based κ).

7 Script interface

File: `jobs/model1_coupled_oscillators/spectrum_model1.py`

- Edit the **PARAMETERS** block at the bottom of the script ($\omega_p, \gamma_p, f_p, \omega_m, \gamma_m, f_m, \kappa$, grid, optional PlasMol CSV overlay).
- Run: `python spectrum_model1.py`
- Outputs: `spectrum_model1.csv`, `spectrum_model1.png`

Columns in the CSV: Frequency, Absorption (normalized), `Re_alpha_m_eff`, `Im_alpha_m_eff`.

8 Limitations

- κ is phenomenological; it does not know the sphere radius or gap. Use Model 2 for geometry-explicit coupling.
- Linear, classical, Markovian damping only—no quantum saturation, no non-Markovian continuum of the metal beyond a single Lorentzian plasmon mode.
- Absolute oscillator strengths f_p, f_m are relative; only ratios and κ set the hybrid lineshape after peak normalization.
- PlasMol’s RT-TDDFT Na line and Meep Au dielectric are richer than two Lorentzians; agreement is expected at the level of peak count, ordering, and rough shifts, not spectroscopic identity.

9 References (entry points)

- Classical coupled-oscillator / plexciton models for plasmon–exciton hybrids (e.g. Wu *et al.*, *Nature* **2010**; subsequent plasmon–exciton reviews).
- Novotny, “Strong coupling, energy splitting, and level crossings” style treatments of two damped oscillators.
- PlasMol Fourier methodology notes (`doc/latex/fourier_driver_methodology.tex`): definition of μ/E_{inc} deconvolution.